



Healthcare Operations Benchmarking Case Study

Anonymized data workflow, benchmarking, and risk-monitoring example

A portfolio example showing how structured data rules and benchmarking can support operational review and decision-making.

THE DOMAS GROUP LLC

Objectives

- Provide actionable insights and flag communities needing closer observation.
- Establish analytic benchmarks and hypotheses to guide ongoing risk monitoring and improvement.

Data Overview

Dataset	Type	Records
Weights	Time-series	De-identified event data
Falls	Event-based	De-identified incident data

Key Business Rules Applied:

- Each weight record represents a single, valid measurement per resident per day.
- Implausible weights (< 70 lbs / > 450 lbs) were excluded to ensure realistic clinical ranges.
- Only the most recent weight per day was retained for accurate trend modeling.
- Monitoring intervals and threshold events ($\pm 5\%$ / $\pm 10\%$) were calculated consistently across communities.
- Fall-related incidents were isolated using a standardized 'is_fall' flag.
- Days-between events were calculated per resident, then summarized using resident-median $\rightarrow$ global median.
- Repeat-fall rates were based on the minimum interval between consecutive falls.
- Community-level rates (e.g., incidents per 100 residents) were normalized using unique resident counts.

Benchmark Discovery - Weight

	Benchmark	What It Captures	Why It Matters
1	% of Residents with ≥ 1 Weight-Change Event ($\geq \pm 5\%$ / 30 d or $\pm 10\%$ / 180 d)	Prevalence of residents showing clinically meaningful change in body weight.	Signals the proportion of residents potentially at nutritional or health risk; high values may indicate unstable populations or measurement inconsistency.
2	Weight-Change Events per 100 Residents	Frequency of threshold-crossing events normalized to population size.	Allows comparison across communities regardless of census; rising rates suggest recurring instability among monitored residents.
3	Median Days Between Weigh-Ins	Typical cadence of resident weight monitoring.	Reflects adherence to expected monitoring schedules; longer intervals reduce ability to detect early risk.
4	% Residents with > 30 Days Between Weigh-Ins	Share of residents exceeding recommended weigh-in frequency.	Highlights monitoring gaps that may delay detection of weight-related health decline or safety risk.

Benchmark Discovery - Weight

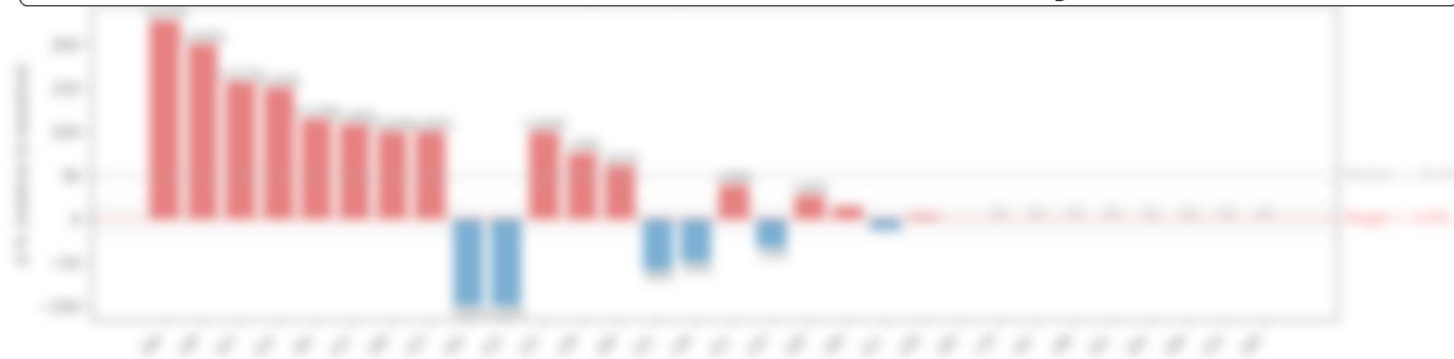
	Benchmark	What It Captures	Why It Matters	Current State
1	% of Residents with ≥ 1 Weight-Change Event ($\geq \pm 5\%$ / 30 d or $\pm 10\%$ / 180 d)	Prevalence of residents showing clinically meaningful change in body weight.	Signals the proportion of residents potentially at nutritional or health risk; high values may indicate unstable populations or measurement inconsistency.	Redacted
2	Weight-Change Events per 100 Residents	Frequency of threshold-crossing events normalized to population size.	Allows comparison across communities regardless of census; rising rates suggest recurring instability among monitored residents.	Redacted
3	Median Days Between Weigh-Ins	Typical cadence of resident weight monitoring.	Reflects adherence to expected monitoring schedules; longer intervals reduce ability to detect early risk.	Redacted
4	% Residents with > 30 Days Between Weigh-Ins	Share of residents exceeding recommended weigh-in frequency.	Highlights monitoring gaps that may delay detection of weight-related health decline or safety risk.	Redacted

Benchmark 1

A subset of communities showed increases in the proportion of residents with clinically meaningful weight-change events, flagging areas for closer monitoring and data-quality review.

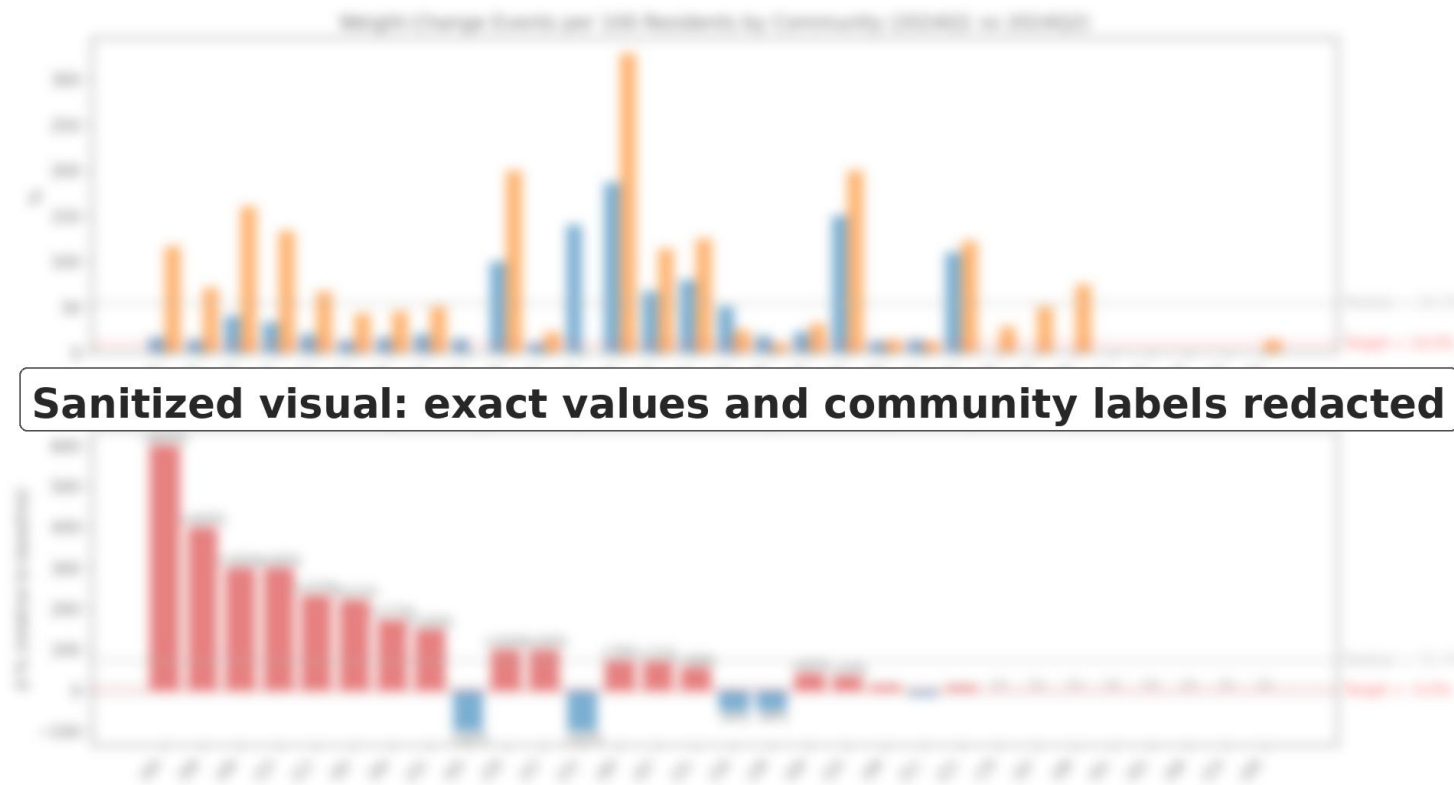


Sanitized visual: exact values and community labels redacted



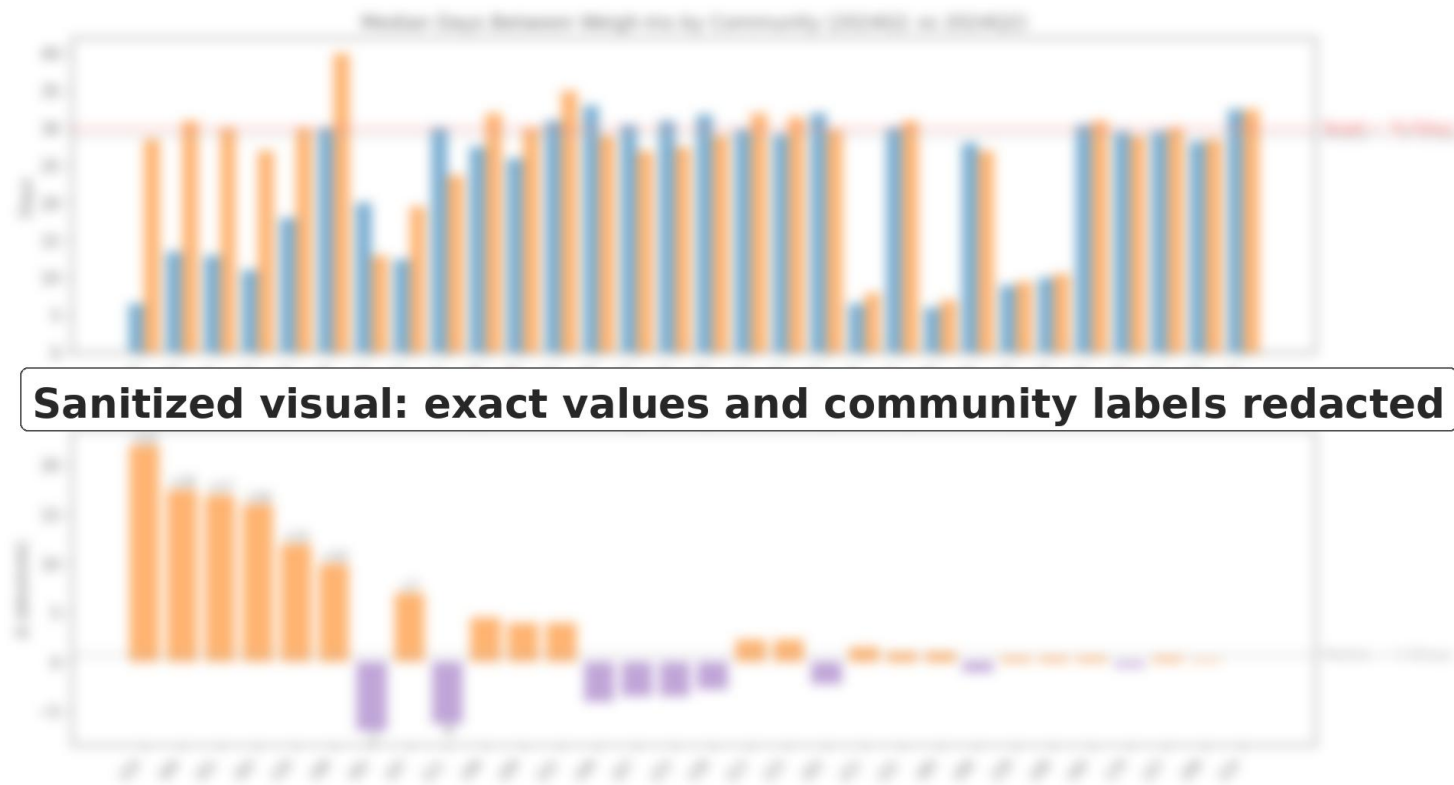
Benchmark 2

Event frequency varied across communities after normalizing by resident count, helping distinguish broad resident involvement from repeated events among smaller groups of residents.



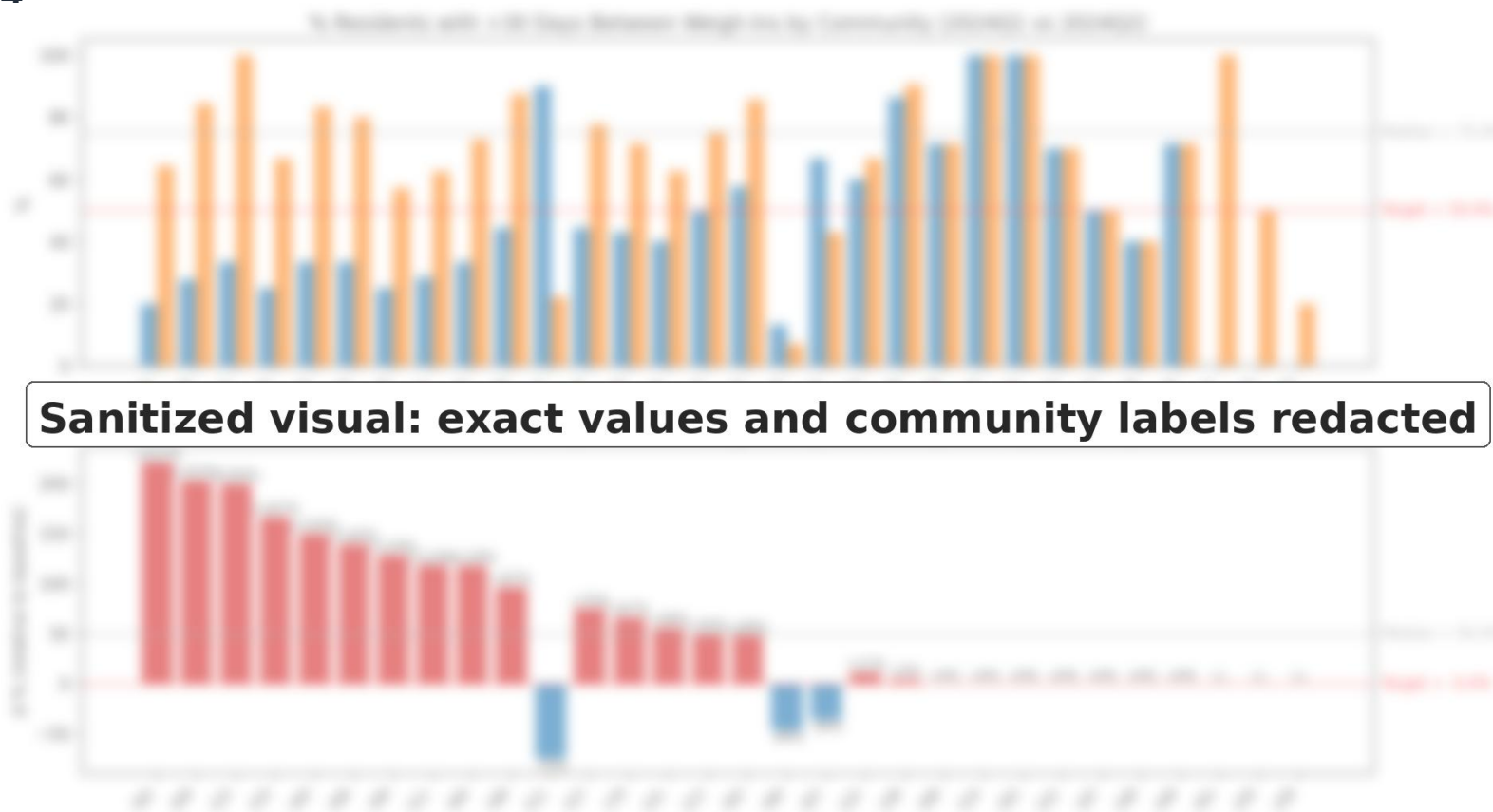
Benchmark 3

Several communities showed longer intervals between weigh-ins, potentially delaying detection of early weight changes and weakening routine monitoring cadence.



Benchmark 4

Many communities had residents exceeding the expected monthly weigh-in cadence, underscoring the risk of missed early warning signs when monitoring frequency declines.



Benchmark Discovery - Incidents

	Benchmark	What It Captures	Why It Matters
1	% Residents with Repeat Fall <30 Days	Share of residents falling again within 30 days.	Rapid recurrence indicator; mirrors readmission-style quality triggers.
2	Falls per 100 Residents	Frequency of fall incidents normalized to population size.	Core safety metric; rising rates flag environments or workflows needing review.
3	Median Days Between Falls	Typical interval between repeat falls (among residents with >1).	Shorter intervals imply unresolved risk or insufficient recovery after events.
4	Fall-to-Weight-Event Ratio	Relationship between fall frequency and weight-change events.	Tests whether physiological instability precedes adverse safety events.

Benchmark Discovery - Incidents

	Benchmark	What It Captures	Why It Matters	Current State
5	% Residents with Repeat Fall <30 Days	Share of residents falling again within 30 days.	Rapid recurrence indicator; mirrors readmission-style quality triggers.	Redacted
6	Falls per 100 Residents	Frequency of fall incidents normalized to population size.	Core safety metric; rising rates flag environments or workflows needing review.	Redacted
7	Median Days Between Falls	Typical interval between repeat falls (among residents with >1).	Shorter intervals imply unresolved risk or insufficient recovery after events.	Redacted

Benchmark 7

Repeat falls increased across many communities, with several showing notable spikes that warrant closer review.



Benchmark 8

Incidents per 100 residents increased in some communities, while most remained relatively stable or decreased.



Benchmark 9

Median days between incidents increased across most communities, with some showing substantial increases in time between events.



Recommendations

To improve weight monitoring compliance and reduce fall risk, we recommend prioritizing communities that show multiple signals across weight and incident benchmarks, such as:

- High % of residents with ≥ 1 weight-event or fall incident, or high events per 100 residents
- Median weigh-in intervals > 30 days or $> 30\%$ of residents exceeding the 30-day weigh-in threshold.
- High repeat-fall rates (< 30 days) or sharp decreases in median days between incidents

Additional Hypotheses & Analytic Opportunities

1. Hypothesis: Communities that weigh residents or document incidents less frequently may appear “stable” simply because events aren’t recorded.

Confirm the true resident denominator per quarter to assess missingness, under-reporting, and census-driven distortion of event rates.

2. Hypothesis: Communities with inconsistent weigh-in patterns may also show higher fall rates or shorter intervals between incidents.

Assess communities with unusually low measurement frequency as potential under-detectors rather than low-risk sites.

3. Hypothesis: Spikes in events may align with specific operational changes, staffing patterns, or seasonal/holiday periods.

Analyze whether spikes in incidents or weight events align with staffing changes, new processes, or seasonal factors.

4. Hypothesis: Weight changes and fall risk vary by BMI range, hydration status, diagnoses, or acuity scores rather than operational issues.

Incorporate height/BMI, hydration-related flags, diagnoses, or acuity indicators to test whether variation reflects case-mix rather than operational inconsistency.



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